Fundamentals Chapter 6

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1. Assuming additivity and homogeneity, we see that $L(c\mathbf{u}+\mathbf{v}) = L(c\mathbf{u}) + L(\mathbf{v}) = cL(\mathbf{u}) + L(\mathbf{v})$. Now, we assume that $L(c\mathbf{u}+\mathbf{v}) = cL(\mathbf{u}) + L(\mathbf{v})$. Let $c = 1$, then $L(\mathbf{u}+\mathbf{v}) = L(\mathbf{u}) + L(\mathbf{v})$. Let $\mathbf{v} = 0$, then $L(c\mathbf{u}) = cL(\mathbf{u})$.

2. If $L(V) = \{\mathbf{0}\}$ then $L(B) = \{\mathbf{0}\}$, so $L(B)$ spans $L(V)$. Furthermore, $\{\mathbf{0}\}$ is a basis for $\{\mathbf{0}\}$, so $L(B)$ is a basis for $L(V)$.

3. Since all vector spaces are isomorphic to $\mathbb{R}^n$ for some n, we can solve this problem for $\mathbb{R}^n$. A basis for $\mathbb{R}^n$ would be a list of $n-1$ zeroes and a 1 in some order. If we take any two of these, say $\mathbf{u}$ and $\mathbf{v}$, it is never the case that $c\mathbf{u}+\mathbf{v} = \mathbf{0}$ because they are linearly independent.

Define the map $L : \mathbb{R}^n \to \mathbb{R}^n$ as $L(\mathbf{u}) = \mathbf{u}$ for all $\mathbf{u} \neq \mathbf{0}$ and $L(\mathbf{0}) \neq \mathbf{0}$. Then for all $\mathbf{u}, \mathbf{v}$ in the basis mentioned earlier, $c\mathbf{u}+\mathbf{v} \neq \mathbf{0}$ so

$$L(c\mathbf{u}+\mathbf{v}) = c\mathbf{u}+\mathbf{v} = cL(\mathbf{u}) + L(\mathbf{v}).$$

Now, let $\mathbf{u} = (1, \ldots, 1)$ and $\mathbf{v} = (-1, \ldots, -1)$. Then

$$L(1 \cdot \mathbf{u}+\mathbf{v}) = L(\mathbf{0}) \neq \mathbf{0} = 1 \cdot L(\mathbf{u}) + L(\mathbf{v}).$$

4. Let $\mathbf{u}, \mathbf{v} \in V$ be vectors. We know that $\mathbf{u} = \sum_{i=1}^{n} a_i v_i$ and $\mathbf{v} = \sum_{i=1}^{n} b_i v_i$ for some constants $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$. So we have that

$$L(c\mathbf{u}+\mathbf{v}) = L\left(\sum_{i=1}^{n} ca_i\mathbf{v}_i + \sum_{i=1}^{n} b_i\mathbf{v}_i\right) = L\left(\sum_{i=1}^{n}(ca_i+b_i)\mathbf{v}_i\right)$$

$$= \sum_{i=1}^{n}(ca_i+b_i)L(\mathbf{v}_i) = \sum_{i=1}^{n} cL(a_i\mathbf{v}_i) + L(b_i\mathbf{v}_i) = cL(\mathbf{u}) + L(\mathbf{v}),$$

which means L is a linear map.

5. Let $V_1 = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ so that $\mathbf{v}_1 + \cdots + \mathbf{v}_n \neq 0$. Then $L(\mathbf{v}_1 + \cdots + \mathbf{v}_n) \neq L(0) = 0$, and also $L(\mathbf{v}_1 + \cdots + \mathbf{v}_n) = L(\mathbf{v}_1) + \cdots + L(\mathbf{v}_n) \neq 0$. So $\{L(\mathbf{v}_1), \ldots L(\mathbf{v}_n)\} = L(V_1)$ is also linearly independent.

6. Consider two fields and an isomorphism $\phi : \mathbb{F}_1 \to \mathbb{F}_2$. Let $K = \ker(\mathbb{F}_1)$, so that $\phi(k) = 0$ for all $k \in K$. Now, assume K is a subfield of $\mathbb{F}_1$. Then 1 is an element in K. so $\phi(1) = 0$. But we see that $1 \cdot \phi(x) = \phi(1 \cdot x) = \phi(1)\phi(x)$, so $\phi(1) = 1$. This is a contradiction, so the kernel is not a subfield of $\mathbb{F}_1$. This means we cannot take the quotient of $\mathbb{F}_1$ and K, so there is no field equivalent of the first isomorphism theorem.

8. We see that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ and $\mathbb{Q}(\sqrt{3}) = \{a + b\sqrt{3} : a, b \in \mathbb{Q}\}$. We can write any element of $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{3})$ uniquely as the ordered pair (a, b). This gives us an isomorphism.

Assume there is an isomorphism $\phi : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{3})$. We know that $\phi(1) = 1$, which means $\phi(q) = q$ for all $q \in \mathbb{Q}$.

Now, consider a polynomial $p(x) = \sum_{i=0}^{n} a_i x^n$. We see that

$$\phi(p(x)) = \phi\left(\sum_{i=0}^{n} a_i x^n\right) = \sum_{i=0}^{n} a_i(\phi(x))^n = p(\phi(x)).$$

Now, assume $p(r) = 0$. This means $\phi(p(r)) = \phi(0) = 0$, so $p(\phi(r)) = 0$. This means if r is a root of $p(x)$ then $\phi(r)$ is too. So ϕ permutes the roots of $p(x)$.

Applying this to $\mathbb{Q}(\sqrt{2})$, we see that $\sqrt{2}$ is a root of the polynomial $x^2 - 2 = 0$. So $\phi(\sqrt{2})$ is also a root of $x^2 - 2 = 0$, so it is $\sqrt{2}$ or $-\sqrt{2}$.

We know that $\sqrt{3}$ must get mapped to by some element of $\mathbb{Q}(\sqrt{2})$, say $a + b\sqrt{2}$. We see that $\phi(a + b\sqrt{2}) = \phi(a) + \phi(b\sqrt{2}) = a + b\phi(\sqrt{2})$. We have that $\phi(\sqrt{2}) = \pm\sqrt{2}$, so $\phi(a + b\sqrt{2}) = a \pm b\sqrt{2}$ where $a, b \in \mathbb{Q}$, which cannot be equal to $\sqrt{3}$. This is a contradiction, so there doesn't exist a field isomorphism from $\mathbb{Q}(\sqrt{2})$ to $\mathbb{Q}(\sqrt{3})$.

9. To show that homogeneity implies additivity, we see that

$$L(x + y) = L\left(x\left(1 + \frac{y}{x}\right)\right) = \left(1 + \frac{y}{x}\right)L(x) = L(x) + \frac{y}{x}L(x)$$

$$= L(x) + \frac{y}{x}(xL(1)) = L(x) + yL(1) = L(x) + L(y).$$

Additivity does not imply homogeneity. To see this, consider a function L where $L(1) = 1$ and $L(\sqrt{2}) = \sqrt{3}$. Then $L(\sqrt{2} + 1) = L(\sqrt{2}) + L(1) = \sqrt{3} + 1$. But

$$\frac{\sqrt{2} + 1}{\sqrt{2}}L(\sqrt{2}) = \sqrt{3}\left(\frac{\sqrt{2} + 1}{\sqrt{2}}\right) \neq \sqrt{3} + 1,$$

so we see that

$$L(\sqrt{2} + 1) \neq \frac{\sqrt{2} + 1}{\sqrt{2}}L(\sqrt{2})$$

which is a counterexample to homogeneity.

11. I'm guessing that the infinite dimensional version is that

$$|V| = |\ker(L) \cup L(V)|.$$

I couldn't figure out how to prove this though.